

4	(a) (i)	two sets of co-ordinates taken to determine a constant value (F/x)	M1	
		F/x constant hence obeys Hooke's law	A1	[2]
		or gradient calculated and one point on line used to show no intercept hence obeys Hooke's law	(M1) (A1)	
	(ii)	gradient or one point on line used e.g. $4.5/1.8 \times 10^{-2}$	C1	
		($k =$) 250 N m^{-1}	A1	[2]
	(iii)	work done or $E_p =$ area under graph or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$	C1	
		$= 0.5 \times 4.5 \times 1.8 \times 10^{-2}$ or $0.5 \times 250 \times (1.8 \times 10^{-2})^2$	C1	
		$= 0.041$ (0.0405) J	A1	[3]
	(b)	$KE = \frac{1}{2}mv^2$		
		$\frac{1}{2}mv^2 = 0.0405$ or $KE = 0.0405$ (J)	C1	
		($v = [2 \times 0.0405/1.7]^{1/2} =$) 0.22 (0.218) m s^{-1}	A1	[2]